# Docker: The Platform for New Generation Virtualisation

This article sheds some light on why traditional virtualisation is losing its sheen and Docker is so popular today.

Virtualisation refers to the process of creating a virtual environment so that minimum resources are used for better control of systems. The main infrastructure needed for virtualisation includes hardware, operating systems, storage devices and a network environment. To build a virtual system, virtualisation technology employs the use of software that mimics hardware capabilities. A single server may run several operating systems, many virtual systems, and numerous applications. Greater efficiency and minimum resources are the key advantages of virtualisation.

Operating system (OS) virtualisation is the process of running several operating systems on a single piece of hardware using virtualisation software. Its usage can help save money for developers and enterprise managers by not squandering costly computing power. As a result of the abstraction provided by virtualisation, hardware resources may be shared across different workloads. These workloads generally co-exist on shared virtualised infrastructure while being completely isolated from one another, allowing mobility across infrastructures as and when required.

The benefits that enterprises receive from virtualisation are quite significant. In terms of capital and operational efficiency, it allows for better server utilisation and consolidation, resource provisioning allocation and utilisation, workload separation, security, and automation. It is also now feasible to provide services and resources in a hybrid cloud, on-premises or in the cloud using virtualisation and software-defined orchestration.

The key advantages of virtualisation technology are:

- Low cost with minimum physical infrastructure
- Easy and fast deployment
- Fault tolerance and disaster management
- Fast backup and recovery
- Improved resource utilisation
- Server consolidation
- Dynamic load balancing
- Virtual infrastructure
- Improved security and system reliability

Use cases of virtualisation technology include:

- Performance sensitive workload management
- Hybrid and multi-hypervisor environment
- Technical workstations
- Virtual data warehouses
- Virtual data lakes
- 5G in telecom sector
- Data warehouse offloading
- Real-time data analytics and dynamic load balancing

## Traditional versus new generation virtualisation

Traditional virtualisation is dependent on separate virtual machines and the installation of operating systems on each of them. The drawback here is that there is a huge load on the host operating system, and lots of computing resources and infrastructure are required. However, for new generation virtualisation using Docker, there is no need to create a separate virtual machine for each operating system. Docker images are available for

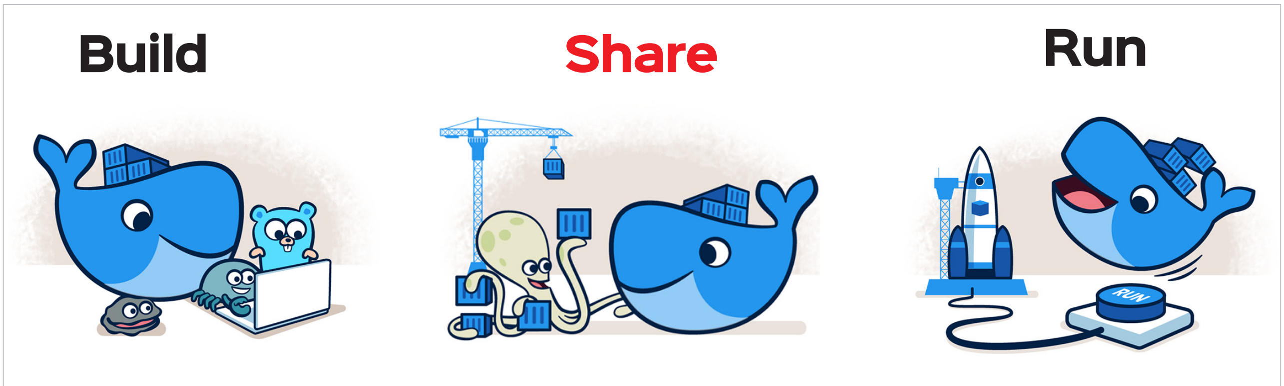


Figure 1: Docker can build, share and run apps anywhere